

HScore is an internal decision criterion, not a standardized external benchmark. Percentage-form components must be represented on a consistent scale before combination.

For split-validity analysis, GroupOverlap is defined by (D.1) at specimen-group level. RTMDet recall values in the added comparator study were computed under the recorded RTMDet evaluation at IoU 0.50 and confidence 0.25, whereas YOLO mask recall is the Ultralytics-exported recall. These definitions are not treated as numerically identical. MMEngine evaluator timing is likewise not reported as pure RTMDet inference latency.

Appendix H - Corruption Protocol Inventory

This appendix keeps classification and segmentation corruption evidence separate. The tasks used different transform families, sample scopes, and reporting rules.

Table H.1. Classification corruption inventory

Family	Retained severity evidence	Reporting boundary
Impulse noise	Clean plus severity levels 1-3 in qualitative panels	Aggregate Macro-F1 and Kappa retained; labels preserved
Gaussian noise	Clean plus severity levels 1-3 in qualitative panels	Aggregate Macro-F1 and Kappa retained; not interchangeable with blur
Contrast reduction	Clean plus severity levels 1-3 in qualitative panels	Aggregate Macro-F1 and Kappa retained
Defocus blur	Clean plus severity levels 1-3 in qualitative panels	Aggregate Macro-F1 and Kappa retained
Low light	Clean plus severity levels 1-3 in qualitative panels	Aggregate Macro-F1 and Kappa retained

The exact transform parameters remain tied to archived transformation definitions rather than inferred from library defaults. The retained aggregate table does not provide a complete per-severity machine-readable ledger.

Table H.2. Segmentation corruption inventory and naming boundary

Family	Retained evidence	Interpretation
Color cast	Quantitative positive-gain row and qualitative examples	One of five selected largest positive gains
Turbidity	Quantitative positive-gain row and qualitative examples	Synthetic visibility and color attenuation
JPEG artifact	Quantitative positive-gain row and qualitative examples	Compression-specific stress condition
Overexposure	Quantitative positive-gain row and qualitative examples	Intensity clipping / illumination stress
Gaussian noise	Quantitative positive-gain row	Distinct from Gaussian blur
Gaussian blur	Qualitative panel only	No quantitative value may be substituted from the Gaussian-noise row